

Aim: To implement an ANOVA Test**Theory:****Q1. What is ANOVA? (Analysis of Variance)**

A statistical test that compares the means of **three or more groups** to see if at least one is different. It works by comparing variance *between* groups to variance *within* groups.

Q2. What is the F-test?

The specific test statistic used in ANOVA. It's a ratio:

$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}}$$

A large F value (much greater than 1) suggests group means are not equal.

Q3. What is the relationship between ANOVA and linear regression?

They are mathematically identical. ANOVA is a special case of linear regression where the predictor variables are categorical (e.g., treatment groups).

Running ANOVA on group means gives the same p-value as running regression with dummy-coded group variables.

Q4. Write a Python program to implement an ANOVA test.**Code:**

```
import scipy.stats as stats

group1 = [21, 24, 22, 23, 25]
group2 = [28, 27, 30, 29, 31]
group3 = [18, 20, 19, 22, 21]

f_stat, p_value = stats.f_oneway(group1, group2, group3)

print(f"F-statistic: {f_stat:.4f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject H0: At least one group mean is different.")
else:
    print("Fail to reject H0: No significant difference between group means.")
```

Output:

```
spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/AIDS-1/Experiments/EXP10 (master)
$ py app.py
F-statistic: 42.0000
P-value: 0.0000
Reject H0: At least one group mean is different.
```

Conclusion:

Successfully implemented Linear Regression in Python.